Abdullah Shamail

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Education

Iowa State University — Ames, IA
Ph.D. Computer Engineering (GPA: 3.90/4.00) — *Expected Aug 2026*
- **Focus:** Scalable Spatio-Temporal Data Management and pattern mining in distributed environments.
- **Honors:** NSF Student Travel Grant Recipient; Preparing Future Faculty Associate (2025).

Iowa State University — Ames, IA
M.S. Computer Engineering (GPA: 3.89/4.00) — *Dec 2024*

Lahore University of Management Sciences — Lahore, Pakistan
B.S. Computer Science — *May 2020*

Experience

Altair Engineering Inc. — Troy, MI
Engineering Data Science (Machine Learning) Intern — *May 2025 – Aug 2025*
- Engineered GNN and Transformer-based architectures for PhysicsAI, achieving a **10× reduction in MAE** and enabling longer-horizon predictions for large-scale physical simulations.
- Deployed scalable ML pipelines using **PyTorch and TensorFlow**, supporting end-to-end training and **GPU-accelerated inference** integrated with production platforms.

Iowa State University — Ames, IA
Graduate Research Assistant (PhD Student) — *Jan 2021 – Present*
- Designed the SDC framework using **transformers**, indexing and spatial partitioning, achieving up to **65× speedups** in diversity-aware clustering for dynamic spatio-temporal datasets. *Submitted 2026.*
- Engineered BARMC, a high-performance framework achieving an **80% speedup** in hydrogen-bond detection for 800k-frame simulations by implementing a lossless spatial pruning algorithm that removed **90% of redundant data**, enabling precise tracking of bond persistence and molecular stability. *Accepted in ACM SIGSPATIAL 2024 and TSAS 2026 (Extension).*
- Developed a domain-specific trajectory compression algorithm that achieved an **80% reduction in storage** and a **3× speedup** in data retrieval for MD simulations. *Accepted in ADBIS 2022 and Information Systems 2024 (Extension).*
- Optimized large-scale text analytics using **Hadoop MapReduce**, resulting in a **2× increase in processing throughput** for bigram frequency identification.

Taar Consulting — Lahore, Pakistan
Software Developer — *May 2018 – June 2018*
- Implemented a Customer Feedback Plugin for Retail Pro Prism, integrating with production APIs to process data from **1,000+ daily transactions**.

Technical Projects

F1 Telemetry Dashboard | ***C#, Blazor WASM, Python, Azure*** — 2026
- Built a full-stack dashboard to visualize high-velocity F1 telemetry, utilizing **Blazor WebAssembly** for high-performance client-side rendering with zero server latency.

Resume Tailor – AI-Powered ATS Optimizer — 2026
- Developed a **client-side web application** leveraging Groq's LLaMA API for dynamic resume tailoring, implementing intelligent keyword optimization and content reordering based on job descriptions.

AutoJobScout – Agentic AI & RAG Platform | ***LangGraph, RAG, LLMs, Python*** — 2025
- Architected a multi-agent workflow using **LangGraph**, achieving **92% matching accuracy** via semantic resume analysis and resume-to-JD alignment.
- Optimized asynchronous microservices to handle 1,000+ daily listings, reducing end-to-end inference latency to **<15 seconds**.

Sarcasm Detection Engine | ***LSTM, SVM, Python*** — 2022
- Trained a deep learning model for news headline classification, achieving **84% precision** and **82% recall** using tokenized dense vector representations.

Technical Skills

Machine Learning & AI: NLP (Transformers), GNNs, CNNs, LSTMs, Generative AI (DCGAN), RAG, Agentic Workflows.
Big Data & Infrastructure: PySpark, Hadoop MapReduce, Apache Flink, Pig Latin, SQL, NoSQL, Linux, GPU Inference.
Frameworks & Tools: PyTorch, TensorFlow, Keras, Scikit-learn, MLflow, LangChain, Git, Docker, Azure, Databricks.
Programming: Python (Expert), C++, Java, JavaScript (ReactJS), MATLAB, C# (Blazor), SQL.